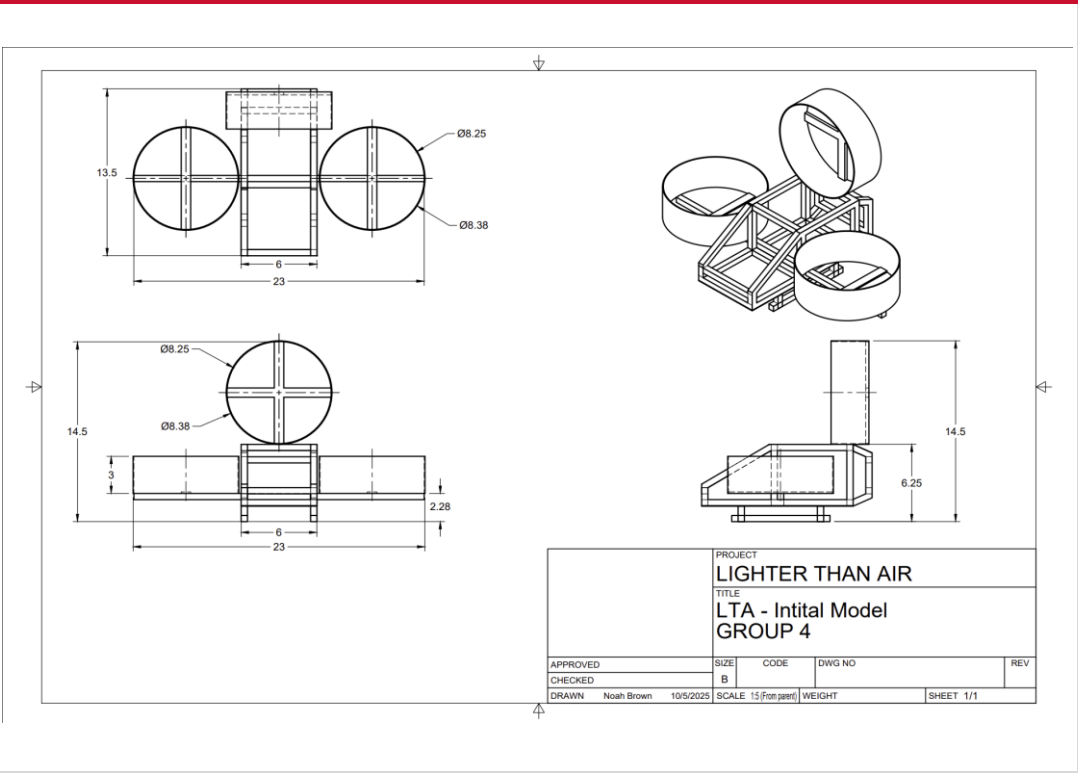


[LTA AIRCRAFT DESIGN]

[Designed and developed a lighter-than-air aircraft from initial concept through physical construction and testing.]



[Final engineering drawing documenting aircraft geometry and dimensions]



[Completed LTA aircraft after construction and testing]

WHAT

[Designed and built a lighter-than-air (LTA) aircraft as part of a 7-person engineering team, integrating a helium balloon, balsa wood structure, propulsion, and flight controls.]

HOW

[Developed the aircraft from initial hand sketches into a 3D CAD model and engineering drawing, then helped translate the design into a physical build while considering weight, propulsion, stability, and controllability.]

RESULT

[Received a qualifying engineering evaluation and achieved a qualifying flight time, allowing our team to advance and compete in a department-wide LTA aircraft competition.]

MY ROLE

[Team Leader & Concept Development/CAD]

TOOLS

[SolidWorks]

TEAM

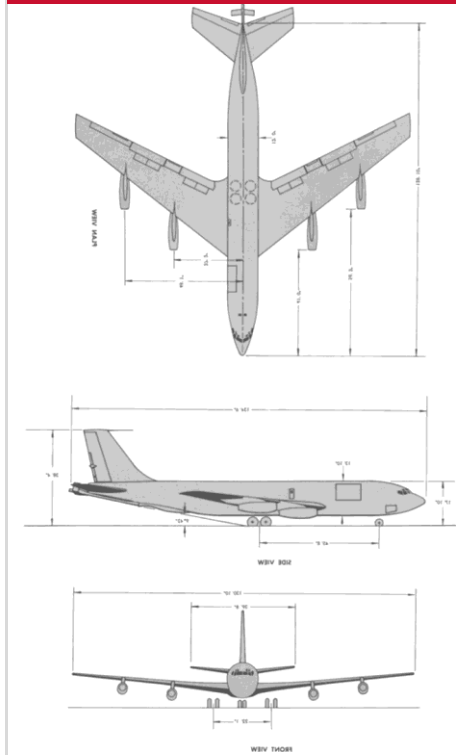
[7-Person Engineering Team]

STATUS

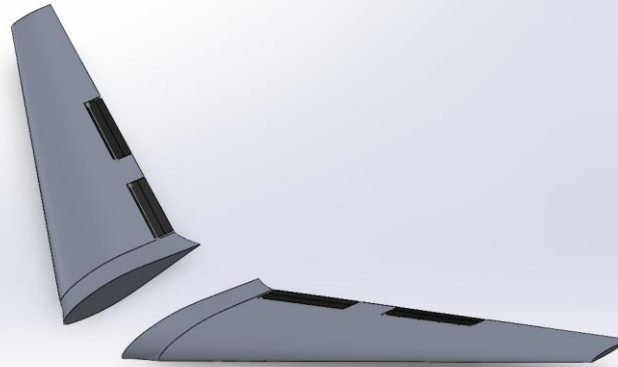
[Complete]

[KC-135 STRATOTANKER]

[Led a five-person aerospace engineering team in developing a full-scale KC-135 Stratotanker CAD model in SolidWorks.]



[KC-135 reference geometry and dimensional data]



[Wing and flap assembly modeled using aircraft airfoil data]



[Completed SolidWorks assembly of the KC-135 Stratotanker]

WHAT

[Developed a detailed 3D CAD model of the KC-135 Stratotanker, including the fuselage, wings, tail, engines, landing gear, and refueling boom.]

HOW

[Led project planning and coordination while modeling the wings and flaps using aircraft dimensions, reference images, and airfoil data. Integrated team components into the final SolidWorks assembly and developed the project presentation and documentation.]

RESULT

[Produced a completed full-scale SolidWorks model and delivered the project's technical presentation and documentation as a team.]

MY ROLE

[Group Leader / Wing & Flap CAD]

TOOLS

[SolidWorks / Airfoil Data]

TEAM

[5-Person Engineering Team]

STATUS

[Complete]

[AIRCRAFT WEIGHT, BALANCE & DENSITY ALTITUDE CALCULATOR]

[Developed MATLAB tools to verify aircraft weight and balance limits and calculate real-time density altitude from live weather data]

Variable	Value
Empty Weight	1,471 lbs
Center of Gravity (Empty)	85.9 in
Front Seats Moment Arm	85.5 in
Fuel Tanks Moment Arm	95 in
Rear Seats Moment Arm	118.1 in
Maximum Ramp Weight	2,400 lbs
Maximum Fuel	50 US Gallons

```
Known Variables
Empty = 1471; % weight in lbs
CGempty = 85.9; % Center of Gravity Empty
FPMomentArm = 85.5; % Front Seat Moment Arm in inches
FPMomentArm = 95; % Fuel Tank Moment Arm in inches
RPMomentArm = 118.1; % Rear Seat Moment Arm in inches
MaxRamp = 2400; % weight in lbs
MaxFuel = 50; % US Gallons

%Extra Credit Part - Asks user if they want to run the calculations
disp('Enter a (0 for Yes) or (1 for No)'); %Displays what to input
answer = input('Would you like to run weight and balance calculations?'); % Prompts to input 0/1 for Y/N
while answer ~=0 %What occurs when input = 0
    fprintf('You have entered 0. The process will now start\n'); % Confirms 0 entered

    [Fuel_User,PilotW_User,CoPilotW_User,PassW_User,Pass2W_User] = User_PromptedInfo(); %Function to gather User Prompted Information for Unknown Variables

    %Fuel Weight Calculations
    Fuel_Weight = FuelW_User.*(4); % Fuel Weight

    %Total Weight Calculations
    People_Weight = PilotW_User + CoPilotW_User + PassW_User + Pass2W_User; % Total People weight
    Ramp_Weight = Fuel_Weight + Empty + People_Weight; % Total Aircraft Weight
    fprintf('The Ramp Weight is %g lbs\n',Ramp_Weight) % Displays Ramp Weight for ease of understanding

    %Checking against Maximum Weight
    if Ramp_Weight > MaxRamp % Error Check - (Unit MISMATCH with AIRCRAFT IS TOO HEAVY)
        disp('Exceeds Maximum weight, Aircraft Will not fly!') % Prints error message
        disp('ERROR in Weight, Do Not Fly Aircraft!') % Prints secondary error message
        disp('Calculations terminated') %Terminates Calculations based on need in problem statement
    else %Otherwise case (Aircraft is under)
        disp('Weight is within range, Aircraft Will be Able to Fly!') %Confirmation message
        disp('Calculations are being executed now')

        %Calculations
        %Moment Calculations
        FrontW_Moment = (PilotW_User + CoPilotW_User) .* FPMomentArm; % Calculates front moment (Pilot + CoPilot Seats)
        FuelW_Moment = Fuel_Weight .* FPMomentArm; % Calculates Fuel Tank Moment (Full Fuel Tanks)
        RearW_Moment = (PassW_User + Pass2W_User) .* RPMomentArm; % Calculates rear moment (Passenger 1 + 2 Seats)
        Empty_Moment = Empty .* CGempty %Given Empty weight and CG, can be used to find the Empty Moment for full calculations

        %Total Moment
        Moment_Total = FrontW_Moment + FuelW_Moment + RearW_Moment + Empty_Moment; % Combines all moments for CG Calculations

        %Center of Gravity Calculation
        CG_Aircraft = (Moment_Total)./(Ramp_Weight); % CG Calculations

        %Acceptable to fly / Not Acceptable to fly
        if CG_Aircraft > 86.8 && CG_Aircraft < 95.8 % CG Parameters
            fprintf('Your center of gravity of %g inches is within parameters\n\n', CG_Aircraft) % Confirmation message for correct CG
            fprintf('Your Ramp Weight is %g lbs, Your Loaded Moment is %f\n', Ramp_Weight,Moment_Total) % Prints Ramp weight and loaded moment as stated in problem statement
            fprintf('It is within Balance and Weight, IT CAN BE FLOWN\n\n') % Confirms it is possible to be flown
        else % Error check (Not flyable)
            fprintf('Your center of gravity of %g inches is not within parameters\n\n', CG_Aircraft) % Error Message
            fprintf('Your Ramp Weight is %g lbs, Your Loaded Moment is %f\n', Ramp_Weight,Moment_Total) % Still prints ramp weight and loaded moment
            fprintf('It is not within Balance and Weight, IT SHOULD NOT Be flown\n\n') % Error Message again - should not be flown
        end % Nested Loop END
    end % If Loop End

    %Prompting For Continuation of Calculations - Extra Credit
    disp('Enter a (0 for Yes) or (1 for No)'); % Displays what to input
    answer = input('Would you like to run weight and balance calculations?'); % Prompts for input
    while answer ~=0
        if answer ==1 %Other condition
            fprintf('You have selected to not run calculations\n'); % Stating calculations are not being run
            disp('Please Try Again when you are ready'); % States a positive message (closing)
        else
            disp('Invalid Input, Try Again!'); %Error Checks if input is entirely wrong - Terminates program
        end
    end
end
```

Enter a (0 for Yes) or (1 for No)
Would you like to run weight and balance calculations 0
You have entered 0, The process will now start

How much Fuel is in the Aircraft (0-50 US Gallons)45
Your Fuel Value of 45 (US Gallons) has been accepted
What is the weight of the pilot (in lbs)200
What is the weight of the copilot (in lbs)125
What is the weight of Passenger 1 (in lbs)80
What is the weight of Passenger 2 (in lbs)100

Thank you for inputting the Information
Running Calculations Now

The Ramp Weight is 2246 lbs
Weight is within range, Aircraft Will Be Able To Fly
Calculations are being executed now
Your center of gravity of 89.5167 inches is within parameters

Your Ramp Weight is 2246 lbs, Your Loaded Moment is 201054.400000
It is within Balance and Weight, IT CAN BE FLOWN

Enter a (0 for Yes) or (1 for No)
Would you like to run weight and balance calculations1

[Aircraft weight and balance parameters used for calculation]

[Weight and balance script and input validation logic in MATLAB]

[Live test case output confirming aircraft is within weight and balance limits]

WHAT

[Built two independent MATLAB calculators: a weight-and-balance checker for a 1965 Piper Cherokee PA-28-180, and a density altitude calculator for Ames Municipal Airport (KAMW) using live API weather data.]

HOW

[Wrote scripts and custom functions with input validation (while loop fuel error checking), calculated ramp weight, moment, and center of gravity against FAA limits, and chained seven equations, including dew point, vapor pressure, virtual temperature, and unit conversions, to compute density altitude from real time temperature, pressure, and humidity.]

RESULT

[Validated both tools across 6 documented test cases (including overweight and out of balance conditions) and cross-checked density altitude output against the NOAA calculator, landing within a ±150 ft margin of error.]

MY ROLE
[Sole Developer / Author]

TOOLS
[MATLAB / OpenWeatherMap API]

TEAM
[Individual Project]

STATUS
[Complete]